ALICE L WALTON SCHOOL OF MEDICINE

# Hereditary Platelet Disorders

Last updated October 30, 2025 16 min read ScholarRx

**Adapted by:** Chelsey Deel
**Editor:** Jordan Finn
Adapted from: Hereditary Platelet Disorders

Listen to this Brick
0:00 / 26:39 1x

Learning Objectives (5) Versions

***After completing this brick, you will be able to:***

1. Identify the most common hereditary platelet disorders: Bernard-Soulier syndrome, Glanzmann thrombasthenia, gray platelet syndrome, and δ-granule deficiency.
2. Differentiate clinical presentations of the most common hereditary platelet disorders.
3. Compare and contrast the pathophysiology and platelet morphology of the most common hereditary platelet disorders.
4. Interpret clinical findings to properly diagnose hereditary platelet disorders.
5. Outline the management plan of hereditary platelet disorders.

**CASE CONNECTION**

MR is a 2-month-old whose parents bring him to the clinic for his 2-month check-up. MR's parents report that their son is eating and sleeping well. "We're a little concerned about his nosebleeds," his mother says. "He gets them three or four times a week. I just hold light pressure for about 5 minutes, and they go away. I remember my brother always had easy bleeding too, and last week, my mom told me that my uncle had some sort of bleeding disorder. We don't have any details on what it was." MR's physical exam is unremarkable.

What is your next best step in evaluating MR's condition? How will you describe this condition to his parents? Consider your answers as you read, and we'll revisit MR at the end of the brick.

GO TO CONCLUSION ↓

## What Are Hereditary Platelet Disorders (LO #1)?

We know that platelets are required to stop bleeding and that they do this by forming clots (Figure 1).

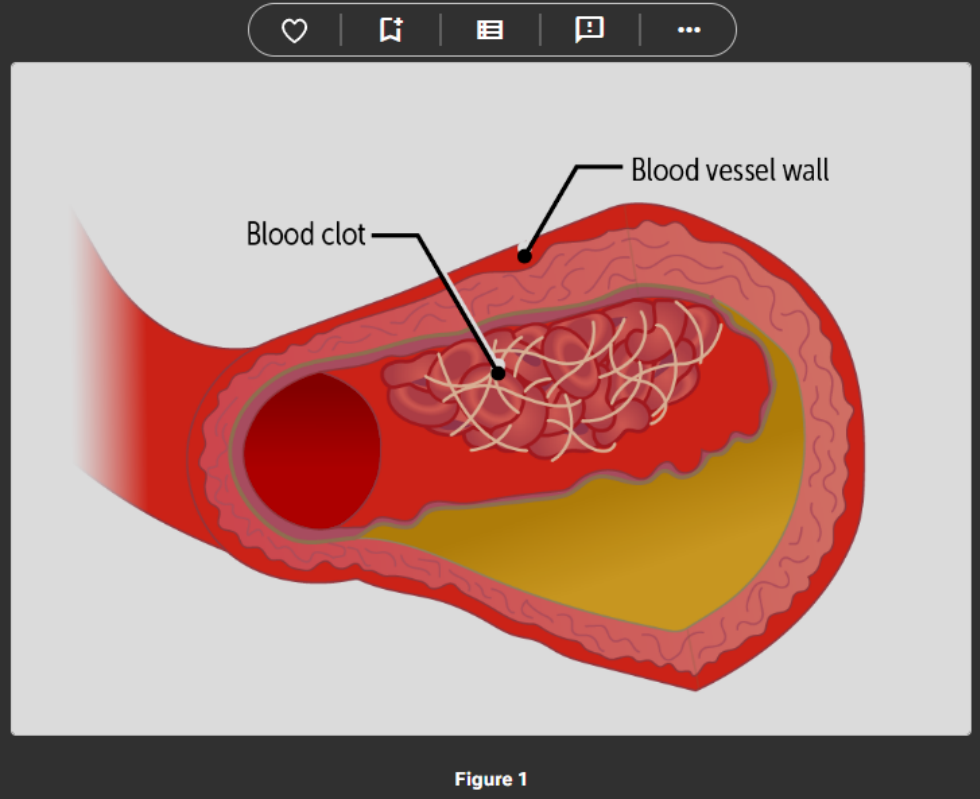


Figure 1

So what happens when the platelets themselves are diseased? The resulting functional disorders can be acquired or inherited; and although the inherited disorders are rare, they do occur.

Among the most common clinically encountered hereditary platelet disorders are those characterized by problems with a platelet's surface receptors or with their granules. These include:

- Bernard-Soulier syndrome
- Glanzmann thrombasthenia
- Gray platelet syndrome
- δ-granule deficiency

In this brick we will take a detailed look at how the hereditary platelet disorders present clinically, how they halt the coagulation cascade, how to differentiate them, and how to treat them.

What are the most common clinically encountered hereditary platelet disorders?

## How Does a Patient with a Hereditary Platelet Disorder Present Clinically (LO #2)?

All of the inherited platelet disorders can present with **mucocutaneous bleeding**. This type of bleeding can be seen on the skin or on mucosal surfaces such as the interior of the mouth, nose, and gums, so patients may note nosebleeds or bleedin[illegible] also may be increased bleeding in the setting of surgery or trauma.

Patients may also notice they have **petechiae**, small, pinpoint breaks in capillaries, seen on the skin surface; **purpura**, which are larger small red/purple lesions; and or **ecchymosis** (purple bruises). Each of these is seen in Figure 2.

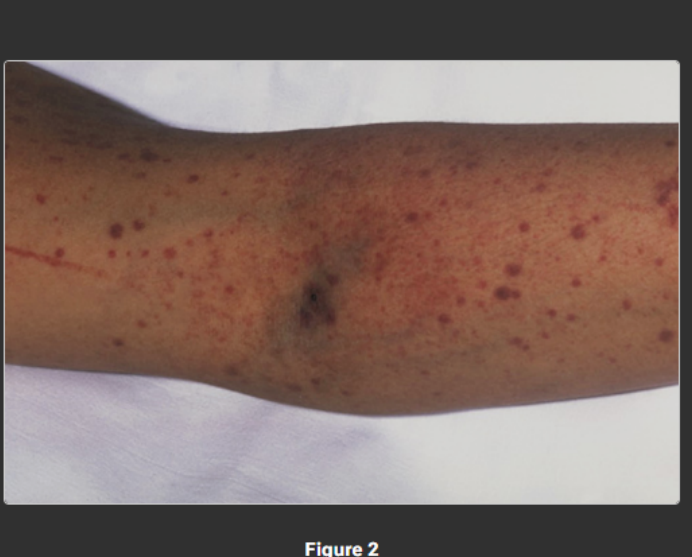


Figure 2

**CLINICAL CORRELATION**

Bleeding that involves deep tissues (muscles) or joints is not a feature commonly associated with platelet disorders and is more common in clotting factor deficiencies like hemophilia.

So, what is uniquely characteristic of each of these inherited disorders?

- **Bernard-Soulier syndrome** may present with severe hemorrhage, coagulopathy, and thrombocytopenia (reduced platelet number).
- **Glanzmann thrombasthenia** in infants often presents at birth with leukocytosis (high white blood cell [WBC] count), delayed umbilical cord separation, or severe bacterial infections, because the disorder often co-occurs with a leukocyte adhesion deficiency (a type of immunodeficiency).
- **Gray platelet syndrome** may show thrombocytopenia.
- **δ-Granule deficiency** may present clinically with recurrent infections because of associated immune deficiency.

What is the common presenting symptom among the hereditary platelet disorders?

## What Is the Pathophysiology of Hereditary Platelet Disorders (LO #3)?

We'll look first at the mechanisms involved in two conditions characterized by platelet receptor problems, then at the mechanisms of two conditions associated with platelet granule problems. But before we detail those mechanisms, let's review how the platelets normally form a plug so that we can recognize when the process fails.

### Normal Platelet Plug Formation

How does a platelet plug form? Recall that this process requires initial adhesion of the platelets to the subendothelium of blood vessels exposed after vessel injury. Platelet activation leads to attachment of platelet GpIb receptors to the complex of von Willebrand factor (vWF) and collagen (Figure 3).

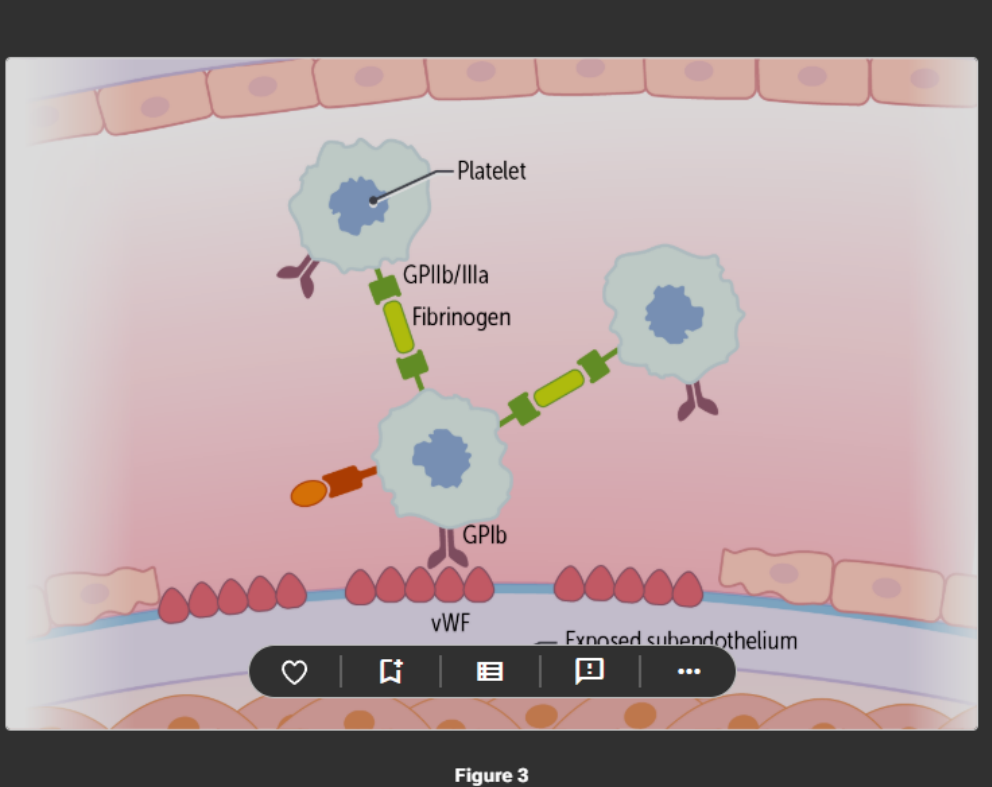


Figure 3

This activation step is followed by platelet aggregation via GpIIb/IIIa receptor binding. Fibrinogen and vWF are released from α granules and further help to form a clot. Delta granules (d-granules) release small molecules such as calcium and serotonin, which are essential for platelet activation. Ultimately, a mesh network of platelets and fibrin is formed, creating the platelet plug and leading to normal coagulation.

### Platelet Receptor Problems

The two most common hereditary platelet disorders are caused by problems with platelet receptors. One is Bernard-Soulier syndrome, and the other is Glanzmann thrombasthenia.

**Bernard-Soulier Syndrome.** This is an autosomal recessive disorder caused by a defect in the glycoprotein Ib (GPIb) receptor. This prevents the platelets from binding to vWF. So, this is a platelet **adhesion** problem. Because these patients cannot form the platelet plug, they can experience severe bleeding. The platelet count is often low.

Histologically, the ***platelets tend to be extremely large*** (giant platelets) on peripheral blood smears. This syndrome is considered one of the giant platelet disorders, a rare group of disorders that includes gray platelet syndrome.

Associated nonplatelet complications include kidney disease (glomerulonephritis) [illegible]ts.

**Glanzmann Thrombasthenia.** This disorder is also autosomal recessive. The abnormalities in this case are in the GPIIb/IIIa receptors, preventing fibrinogen from binding platelets together. So, this is a platelet **aggregation** problem. As a result, even though platelet plug formation will begin, the patient may only get one or two platelets bound to each vWF rather than a cluster of many platelets. This leads to abnormal bleeding.

Histologically, peripheral blood smears will show normal-appearing platelets.

There may be concurrent defects in leukocyte function (e.g., leukocyte adhesion deficiency), leading to leukocytosis, severe bacterial infections, and a history of delayed separation of the umbilical cord at birth.

Table 1.

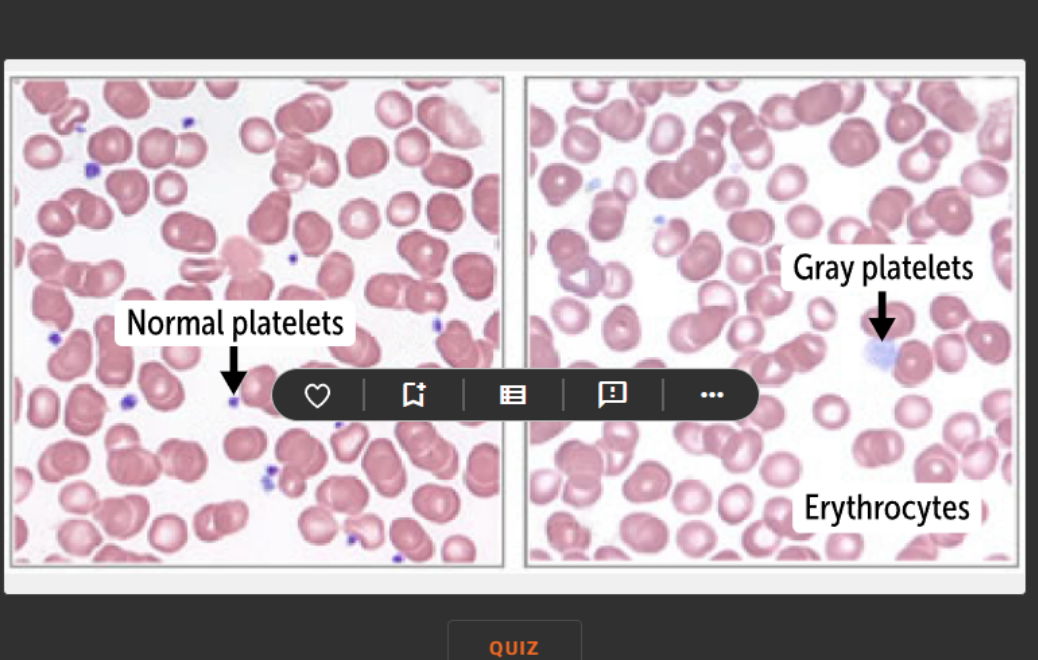
| Characteristic | Bernard Soulier | Glanzmann |
|---|---|---|
| Defect | GP Ib (VWF receptor) | GP IIb/IIIa (Fibrinogen receptor) |
| Inheritance | Autosomal Recessive | Autosomal Recessive |
| Platelet Count | Decreased | Normal |
| Platelet Appearance | Abnormally large | Normal |
| Severity | Variable | Variable |

What is the common platelet defect in both Bernard Soulier syndrome and Glanzmann thrombasthenia?

### Platelet Granule Problems

The other hereditary platelet disorders we will discuss involve **disorders of platelet secretory function** from platelet granules. Examples of these disorders include gray platelet syndrome and δ-granule deficiency.

**Gray Platelet Syndrome.** This syndrome is defined as a reduction (or complete absence) of α granules. Without enough α granules, formation of platelet plugs will be stunted or stopped. The lack of α granules leaves these giant platelets appearing empty, or gray, when compared with more darkly pigmented normal platelets (Figure 4). There can be giant platelets as well.



QUIZ

Figure 4

The cause of this disorder is poorly understood, but multiple genes involved in α granule production have been implicated. Its inheritance may be autosomal recessive, autosomal dominant, or even X-linked, depending on the gene involved.

Other features that can be seen with gray platelet syndrome include progressive thrombocytopenia, myelofibrosis (bone marrow replacement with connective tissue), splenomegaly, and elevated serum $B_{12}$ values.

**δ-Granule Deficiency.** This condition is characterized by a decrease or absence of δ (delta) granules, leading to low secretion of factors needed for clotting. It may also be linked to decreased vWF, stunting the formation of platelet plugs.

The platelets are histologically normal on the blood smear. Electron microscopy can detect the abnormal granules.

This disorder is usually found in conjunction with another hereditary disorder, such as Wiskott-Aldrich syndrome or Chediak-Higashi syndrome (a hereditary cause of immunodeficiency), so patients often have increased infections. Less commonly, it can be s[illegible]

**CLINICAL CORRELATION**

Wiskott-Aldrich syndrome is an X-linked disorder that involves mutations in the *WASP* gene (Wiskott-Aldrich syndrome protein). Signs and symptoms include predisposition to infections, microthrombocytopenia, and eczema.

Features of these disorders are summarized in Table 1.

Which of the inherited platelet disorders are most often associated with multiple infections?

## How Do We Diagnose Hereditary Platelet Disorders (LO #4)?

Now that we've learned about the inherited platelet disorders, how do we differentiate them? After a thorough history and physical exam detecting **mucocutaneous bleeding**, we order laboratory tests.

Patients will have an **abnormal platelet function test**, such as the PFA-100 (platelet function analyzer), which has largely replaced the bleeding time as the best test for platelet function.

**CLINICAL CORRELATION**

Platelet aggregation studies show characteristic patterns for both Bernard-Soulier syndrome and Glanzmann thrombasthenia:

**Bernard-Soulier syndrome:** a normal aggregation response with ADP, arachidonic acid, collagen, and epinephrine with the absence of an aggregation curve with ristocetin.

**Glanzmann thrombasthenia:** no platelet aggregation in response to collagen, ADP, epinephrine, and arachidonic acid. The aggregation response to high-dose ristocetin is usually normal, but may be reversible in some cases.

On the **complete blood count** (CBC), platelet counts can be decreased, normal, or increased. Because platelets are affected and not the coagulation cascade, the prothrombin time (PT) and partial prothromboplastin time (PTT) will be normal.

The **peripheral blood smear** may be helpful:

- Very large platelets are often seen in Bernard-Soulier and gray platelet syndrome.

Which hereditary platelet disorders commonly show very large platelets?

## How Do We Treat Hereditary Platelet Disorders (LO #5)?

Treatment depends on the specific disorder and bleeding severity. In any of the disorders, patients with clinically severe bleeding or low platelets may require platelet or blood transfusion.

However, repeated transfusions may lead to the formation of antiplatelet antibodies (especially against platelet glycoproteins) and HLA antibodies in these patients. These antibodies may limit the use of subsequent transfusions. Patients may also require vaccination against hepatitis B, avoidance of nonsteroidal anti-inflammatory drugs (NSAIDs), correction of iron deficiency, and proper dental hy[illegible]ing.

Some specific treatments include:

- Bernard-Soulier syndrome can be treated with recombinant factor VIIa or a thrombopoietin agonist (eltrombopag).
- Glanzmann thrombasthenia may benefit from hematopoietic stem cell transplants if symptoms are recurrent and severe. High-dose estrogen can also help stop excessive bleeding during menarche.
- Gray platelet syndrome can be treated with thrombopoietin agonists.
- δ-Granule deficiency associated with immune deficiency requires antimicrobial prophylaxis and patients may benefit from GM-CSF (granulocyte macrophage colony-stimulating factor) therapy. δ-Granule deficiency can also be treated with hematopoietic cell transplant.

Why might it be unwise to repeatedly transfuse platelets in patients with hereditary platelet disorders?

**CASE CONNECTION**

BACK TO INTRODUCTION ↑

Thinking back to MR, what is your next best step in evaluating his condition? How do you explain his condition to his parents?

You order a CBC, and all results are normal, including platelet count. PT and PTT are also normal, but the PFA-100 test for abnormal bleeding time is increased. You make a diagnosis of Glanzmann thrombasthenia. You explain to MR's parents that platelets are cells that plug the hole of a damaged blood vessel by clumping together. Their son has an inherited condition in which his platelets are unable to do this. He has enough of them, but they don't do their job very well. You ask MR to visit with a hematologist to discuss potential treatment options and to follow up with you in 2 months.

## Summary

- Hereditary platelet disorders are rare disorders that may cause abnormal bleeding.
- Among the most common are Bernard-Soulier syndrome, Glanzmann thrombasthenia, gray platelet syndrome, and δ-granule deficiency.
- Patients with hereditary platelet disorders present with mucocutaneous bleeding seen as petechiae, purpura, or ecchymosis; or increased bleeding with trauma or surgery.
- Bernard-Soulier syndrome and Glanzmann thrombasthenia are autosomal recessive disorders.
- In gray platelet syndrome and δ-granule deficiency, the inheritance patterns and underlying defects are poorly understood.
- Bernard-Soulier s[illegible]tor, preventing adhesion of platelets to von Willebrand factor (vWF) at the subendothelium.
- Glanzmann thrombasthenia is a defect in the GpIIb/IIIa receptors, preventing formation of the fibrinogen bridges between aggregating platelets.
- Gray platelet syndrome is characterized by decreased or absent platelet α granules, giving the platelets an empty or gray appearance.
- δ-Granule deficiency shows low or absent platelet δ granules and is often associated with other disorders affecting the immune system (e.g., Chediak-Higashi syndrome)
- Basic (but nondiagnostic) tests used to initially diagnose inherited platelet adhesion disorders include platelet count, prothrombin time, and partial thromboplastin time.
- Abnormal platelet function may be assessed with the platelet function analyzer (PFA-100).
- Abnormal platelet function will prompt more specific histologic and genetic testing for diagnosis.
- Platelet transfusion is frequently needed to control spontaneous or severe bleeding.
- Excess platelet transfusion can lead to development of antiplatelet antibodies.
- Eltrombopag may be used in patients with Bernard-Soulier syndrome and gray platelet syndrome.

## Review Questions

**1. A 2-year-old child is brought to the family medicine clinic with notable bleeding within th[illegible]hymoses over the forearms. The mother says the child has had easy bruising and seemingly constant infections throughout her life, as have multiple members of her family. She has never received any blood products but has been hospitalized three times for severe infections. The complete blood count (CBC) and peripheral blood smear show low platelet count, but the platelets appear normal. The PFA-100 test shows slow clotting. Which of the following tests is the best next step in management?**

A. ○ Administer cryoprecipitate
B. ○ Administer packed red blood cells
C. ● Electron microscopy of the peripheral blood
D. ○ Urinalysis to check for hematuria
E. ○ Vitamin $B_{12}$ serum level

**Hide Explanation**

The correct answer is electron microscopy of the peripheral blood (C). The blood smear is examined for δ granules. The history of bleeding and infections in an infant/young child, along with the family history of similar problems, suggests a combined bleeding and immune deficiency disorder, such as Chediak-Higashi syndrome. This may present with mucocutaneous bleeding due to alpha granule deficiency, along with immunodeficiency. Initial labs will commonly show decreased platelet counts, and the PFA-100 bleeding test will be abnormal, as here. The next step in diagnosis is t[illegible]py, which can detect the abnormalities. There is no need for the patient to receive cryoprecipitate, which is used for clotting factor deficiencies, not platelet disorders (A). There is no need for red blood cell (RBC) transfusions, since the complete blood count (CBC) apart from platelet count was normal (B). The patient is not experiencing any urinary symptoms, so a urinalysis to look for hematuria (D) is not correct (however, it would be pertinent to check for additional infections). A vitamin $B_{12}$ serum level (E) is not indicated because there is no macrocytic anemia on the CBC. The level can be increased in gray platelet syndrome.

**2. A young couple presents to your office to ask about the health of their unborn child. The father has Bernard-Soulier syndrome, and the mother's genetic background is unknown. What should you tell the couple about their unborn child's risk of inheriting Bernard-Soulier syndrome?**

A. ○ Bernard-Soulier syndrome has a variable genetic inheritance, so we cannot determine if your child will inherit the disorder, regardless of the genetic background of the parents.
B. ● It is an autosomal recessive disease, so if the mother is a carrier, the child will have a 50% chance of being either a carrier or developing Bernard-Soulier syndrome.
C. ○ It is an autosomal recessive disease, so there is a 50% chance the child will be a carrier of the gene but will not develop Bernard-Soulier syndrome, if the mother is not a carrier of the gene.
D. ○ It is an autosomal recessive disease, so there is a 100% chance the child will have Bernard-Soulier syndrome if the mother is not a carrier.
E. ○ It is an autosomal dominant disease, so there is a 100% chance the child will have Bernard-Soulier syndrome, regardless of the mother's genetic background.

**Hide Explanation**

The correct answer is it is an autosomal recessive disease, so if the mother is a carrier, the child will have a 50% chance of being either a carrier or developing Bernard-Soulier syndrome (B). Bernard-Soulier syndrome is an autosomal recessive disorder characterized by severe mucocutaneous bleeding and altered function of the GPIb receptor on the platelet surface, preventing binding to activated von Willebrand factor (vWF) in injured subendothelium. Since the mode of inheritance is established, variable inheritance is incorrect (A). Because the mode of inheritance is autosomal recessive, two copies of the gene encoding the syndrome will need to be passed on to the child. If the mother is genetically normal (C), the child will only receive one copy from the father; in this case, there is 100% chance the child will be a carrier and 0% chance the child will have Bernard-Soulier syndrome. If the mother is a carrier for Bernard-Soulier syndrome (D), there is 50% chance the child will be a carrier and 50% chance the child will have Bernard-Soulier syndrome. The mode of inheritance is not autosomal dominant (E). However, if it were, only one copy of the gene would be needed to incite the syndrome. And in that case, the child would have 100% chance of having Bernard-Soulier syndrome, regardless of the genetic profile of the mother.

**3. A 1-month-old infant is brought to the pediatrician for an initial outpatient visit. The parents are concerned that the baby bruises easily. The infant was hospitalized for 3 weeks following birth because of several infections. The mother and father have multiple relatives who also bruise easily, and many have received blood products. On exam the umbilical cord has not yet detached from the infant. The routine complete blood count (CBC) shows normal cell counts and normal-appearing platelets. Which of the following disorders is most likely in this patient?**

A. ○ Bernard-Soulier syndrome
B. ○ δ-Granule deficiency
C. ● Glanzmann thrombasthenia
D. ○ Gray platelet syndrome
E. ○ von Willebrand disease

**Hide Explanation**

The correct answer is Glanzmann thrombasthenia (C). The presentation of an infant with easy bruising, recurrent infections, and delayed detachment of the umbilical cord, along with the family history of bleeding and bruising, raises concern for Glanzmann thrombasthenia, an autosomal recessive disorder of the platelet GPIIb and GPIIIa receptors. Bernard-Soulier syndrome (A), gray platelet syndrome (D), and von Willebrand disease (E) all can present with easy bruising but do not show increased infections. δ-Granule deficiency (B) can show combined bleeding and increased infections, but it is not associated with lack of umbilical cord detachment.

**4. What additional treatments may a patient with δ-granule deficiency associated with immune deficiency require for management?**

A. ● Antimicrobial prophylaxis
B. ○ High dos[illegible]
C. ○ Recombinant Factor VIIa
D. ○ Splenectomy
E. ○ Thrombopoietin agonists (i.e. eltrombopag)

**Hide Explanation**

The correct answer is antimicrobial prophylaxis (A). High-dose estrogen (B) can be used to stop excessive bleeding during menarche for patients with Glanzmann thrombasthenia. Recombinant Factor VIIa (C) can be used as a treatment for Bernard-Soulier syndrome. Splenectomy (D) is not a typical treatment for δ-granule deficiency. Thrombopoietin agonists (E) can be used to treat Bernard-Soulier syndrome and gray platelet syndrome.

**How would you rate this Brick?**

## References

- Greer JP, et al, eds. [illegible]d. Wolters Kluwer; 2018.
- Hoffbrand AV, Vyas P, Campo E, et al, eds. *Color atlas of clinical hematology: Molecular and cellular basis of disease.* 5th ed. John Wiley & Sons, Incorporated; 2019
- https://onlinelibrary.wiley.com/doi/10.1111/bjh.15409
- https://www.ncbi.nlm.nih.gov/books/NBK538270/
- https://www.ncbi.nlm.nih.gov/books/NBK557671/
- https://www.uptodate.com/contents/inherited-platelet-function-disorders-ipfds
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Michelson AD, ed. *Platelets*. 4th ed. Academic Press; 2019.

## Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Identify the most common hereditary platelet disorders: Bernard-Soulier syndrome, Glanzmann thrombasthenia, gray platelet syndrome, and δ-granule deficiency.~~
02 ☑ ~~Differentiate clinical presentations of the most common hereditary platelet disorders.~~
03 ☑ ~~Compare and contrast the pathophysiology and platelet morphology of the most common hereditary platelet disorders.~~
04 ☑ ~~Interpret clinical findings to properly diagnose hereditary platelet disorders.~~
05 ☑ ~~Outline the management plan of hereditary platelet disorders.~~

- https://www.ncbi.nlm.nih.gov/books/NBK557671/
- https://www.uptodate.com/contents/inherited-platelet-function-disorders-ipfds
- Kitchens CS, Kessler CM, Konkle BA, et al, eds. *Consultative Hemostasis and Thrombosis*. 4th ed. Elsevier; 2019.
- Michelson AD, ed. *Platelets*. 4th ed. Academic Press; 2019.

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Identify the most common hereditary platelet disorders: Bernard-Soulier syndrome, Glanzmann thrombasthenia, gray platelet syndrome, and δ-granule deficiency.~~
02 ☑ ~~Differentiate clinical presentations of the most common hereditary platelet disorders.~~
03 ☑ ~~Compare and contrast the pathophysiology and platelet morphology of the most common hereditary platelet disorders.~~
04 ☑ ~~Interpret clinical findings to properly diagnose hereditary platelet disorders.~~
05 ☑ ~~Outline the management plan of hereditary platelet disorders.~~

**Objective Confidence: 5/5**